

ECHO: An Efficient Heuristic Viewpoint Determination Method on Frontier-Based Autonomous Exploration for Quadrotors

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Abstract—As a popular drone application, autonomous exploration suffers from low efficiency. To address the issue of repeated and unnecessary exploration, especially in a large-scale and cluttered environment, this letter proposes an efficient heuristic viewpoint determination method on frontier-based autonomous exploration, which includes viewpoint generation, evaluation, and refinement. A Gaussian sampler is employed to randomly generate higher-quality initial viewpoints; meanwhile, a fresh heuristic evaluation function is designed to select the next viewpoint; besides, a refinement strategy is presented to improve the viewpoint. Extensive simulations and real-world tests indicate that the proposed method outperforms the state-of-the-art frontier-based method by 15%-25% in almost all scenarios.

Index Terms—Aerial systems: applications, search and rescue robots, motion and path planning.

I. INTRODUCTION

UNMANNED aerial vehicles (UAV), especially quadrotors, are gaining more and more attention and are used in various applications for their natural agility. Search, inspection, and rescue, with autonomous exploration as their core, are some of the most promising applications. Autonomous exploration, which drives the robot to explore and obtain the environment information gain (EIG) automatically, is being studied by an increasing number of scholars. It was validated and highlighted in the DARPA Subterranean Challenge [1], [2], [3], where multi-robots (legs, wheels, UAV, etc.) were equipped with different sensors to rapidly explore large-scale harsh environments and obtain complete EIG.

The EIG is defined as the amount of new information about the environment that the robot can obtain by moving to a

pose [4]. The studies [5], [6], [7], [8] have introduced different formulations for calculating the value of EIG.

Some research focuses on the reconstruction of objects (maps, buildings, etc.) and pursues the global and consistent accuracy of EIG [6], [9], [10]. However, more research pays attention to the speed of autonomous exploration, i.e., how to drive UAVs to obtain more complete EIG faster [4], [11], [12]. For sufficient EIG, autonomous exploration has evolved into two classical methods: the sampling-based method and the frontier-based method.

The sampling-based method generates a series of candidate viewpoints in known space by a rapidly-exploring random tree (RRT)-like method and picks the next best viewpoint (NBV) [7] by evaluating the EIG. As another mainstream method to autonomous exploration, the frontier-based method generates several frontier clusters first, obtains the viewpoints for each frontier cluster, and determines the NBV for the most EIG. Regardless of the method used, viewpoint decision is the most important module of autonomous exploration and directly determines the exploration efficiency.

However, most methods simply use greedy and single strategies, such as picking the viewpoint with the closest Euclidean distance [8] or the maximum immediate local EIG [11] as the next target. Some work ignores the importance of viewpoint determination, inevitably leading to inefficient and frequently repeated explorations.

Typical and common situations that cause repeated exploration are as follows:

- 1) In a large-scale environment, the boundary space is not thoroughly explored on the first pass, and the agent needs to navigate again after exploring other spaces far apart.
- 2) In a cluttered environment, the complex space surrounded by obstacles and corners is not perceived and valued, and the agent needs to navigate again costly.

To address the above issues, this letter proposes an Efficient Heuristic viewpoint decision method called **ECHO**. As shown in Fig. 1, the proposed method includes viewpoint generation, evaluation, and refinement.

In the proposed method, after creating several frontier clusters, a Gaussian distribution sampler is used to generate sufficient high-quality candidate viewpoints for each cluster and select the single viewpoint in each cluster that covers the most frontier information. Then, a comprehensive and flexible heuristic

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More details of the experiments can be found at <https://youtu.be/Z9dYpEOGXcI>.

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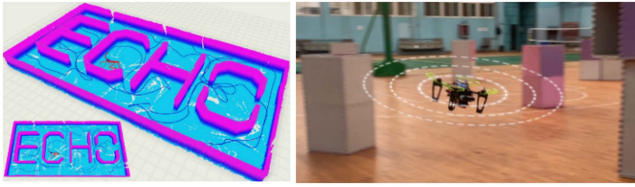


Fig. 1. ECHO in the autonomous exploration.

function is designed to evaluate the cost of each viewpoint, and the best viewpoint is used as the next target. When the refinement is triggered due to the too-close distance between the viewpoint and the current position, the yaw of the viewpoint will be improved for safety and energy-saving. Meanwhile, the quadrotor will be driven to reach the best viewpoint by trajectory planning and will conduct the exploration work iteratively until there is no unknown space.

Our method ECHO is compared with the state-of-the-art frontier-based method FUEL [12] in simulation and real-world experiments. The experimental results show that ECHO can complete the whole exploration efficiently and safely, and improve the exploration time and path by 15~25% with stable efficiency compared to FUEL, regardless of a cluttered indoor quadrotor or a large-scale outdoor quadrotor. In summary, **the contributions of this letter** are:

- 1) A 2D Gaussian sampler is proposed, which randomly generates higher-quality initial candidate viewpoints rapidly by considering the sensor range and environment.
- 2) A fresh and comprehensive heuristic viewpoint evaluation function is designed, which picks the next viewpoint and avoids unnecessarily repeated exploration.
- 3) A flexible viewpoint refinement strategy is exploited, which improves the viewpoint to achieve higher safety and save energy when the special case is triggered.
- 4) Extensive simulations and real-world tests in different scenarios verify the effectiveness of our method.

II. RELATED WORK

A. Fast Autonomous Exploration Method

Fast Autonomous exploration, as mentioned before, is a task that requires the robot to automatically traverse the entire unknown space and obtain complete EIG as fast as possible.

The autonomous exploration work can be mainly divided into sampling-based methods and frontier-based methods. The sampling-based methods determine the next best views (NBVs) [7] by randomly sampling candidate points in a known and accessible space and evaluating each point. To maximize the EIG, RRT-like methods [5], [13] and motion primitives [14] are exploited to sample candidate points with higher quality. However, these methods cannot guarantee complete coverage in large-scale environments and are easy to trap into a local minimum in narrow openings. To solve the above problems, [15], [16] reused the past information to continuously build the roadmaps with memory, and [17] computed the exploration path at two levels: coarsely at the global scale and finely around the

robot. These works are productive and inspiring, but they require certain priors, and their efficiency can be further improved.

The frontier-based methods define the frontier as the boundary between unknown regions and known regions [8] and consider the exploration as completed when there is no frontier in the space, which naturally guarantees the integrity of the map coverage. Some works [20], [21] focus on efficiently establishing frontiers with limited onboard resources, while it was verified that a too-slow frontier search may cause the UAV to repeatedly explore known space [19]. Some works [12], [23] concern motion safety during exploration execution. The work [12] determined the next local target and generated a sufficiently safe polynomial trajectory [24]. The work [23] noticed the EIG in the process of heading to the viewpoints and considered the EIG term during trajectory generation. However, with the rapid development of simultaneous localization and mapping (SLAM) and quadrotor planning technology, the generation and determination of viewpoints, as crucial components of autonomous exploration, have become increasingly important.

B. Viewpoint Generation and Decision

Autonomous exploration can be regarded as a pipeline system that includes environment representation, viewpoint determination, and motion planning. Among them, viewpoint determination (involving generation and decision) is the key link that best characterizes autonomous exploration and most affects efficiency. As mentioned above, quadrotors may repeatedly explore the same space due to the limitation of viewpoint determination. To address this issue, this letter proposes a viewpoint determination method to improve exploration efficiency while ensuring complete coverage. Some previous works have demonstrated the key role of viewpoint determination.

The original frontier-based method [8] greedily selects the frontier with the closest Euclidean distance as the viewpoint. [11] chose the frontier within the field of view (FOV) to minimize the velocity change for a consistently high flight speed on the premise of safety. Also, a frontier-based differentiable measure of information gain [21] was adopted to optimize the paths with gradient information. The above work has improved the determination method to some extent, but these methods can only be applied to specific environments and tasks.

The sampling-based method is more advantageous for viewpoint determination because it flexibly allows various and comprehensive information gain formulation. [5] expanded an RRT with viewpoints and calculated the value including information gain and distance cost. [6] maintained a single tree continuously using a rewiring technique inspired by RRT* [25] and RT-RRT* [26] and introduced a global normalization utility function that considers the gain from the current node and future node. However, the aforementioned sampling-based methods all sample in a limited part of space, resulting in insufficient exploration and a lack of stop criteria.

To combine the advantages of both methods, a growing body of work was proposed. [27] predicted the occupancy of unseen space to generate future frontiers and evaluation viewpoints with predictive information gains. However, similar to another

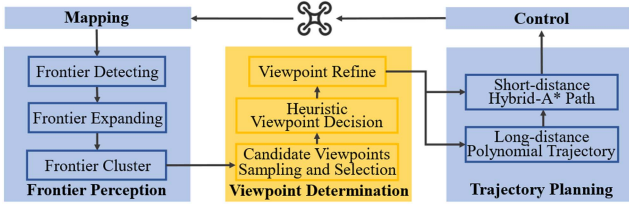


FIG. 2. Overview of the autonomous exploration system proposed in this letter (the yellow box is the focus of this letter).

prediction-based method [28], it depends on the quality of the training data, and the effect is not stable in practice. [20] proposed to sample candidate viewpoints close to frontiers for less sample cost. [12], [29] also sampled viewpoints around frontiers and found the global shortest tour passing through all the viewpoints. However, the uniform sampling method makes most points scattered in the invalid area for EIG. Besides, [12] generated a global tour covering all frontier clusters similar to [29] and modeled it as a traveling salesman problem (TSP) to pick the next viewpoint as the local target. However, the efficiency of TSP to determine the viewpoint has been questioned in recent work [4]. The validation comparison in [4] concluded that the simple exploration heuristic strategy outperforms TSP in most cases. This was also verified in most scenarios by comparing FUEL [12] with the proposed heuristic method **ECHO**, which adopts the frontier baseline for complete coverage and the randomly sampling-based approach for efficient viewpoint determination.

III. SYSTEM OVERVIEW

This letter focuses on the efficient heuristic viewpoint determination method on the frontier-based framework to randomly sample candidate viewpoints and evaluate exploration viewpoints heuristically.

The process is illustrated in Fig. 2. After detecting, expanding, and clustering all frontier information [12], this study adopts a 2D Gaussian distribution sampler to randomly generate enough higher-quality candidate viewpoints in each cluster. The most informative viewpoint is selected to represent the corresponding frontier cluster. Meanwhile, a well-established heuristic cost function is carefully designed to determine the next best viewpoint, which is picked as the local navigation target. Moreover, to handle the case that the next navigation target is too close, this letter also proposes a viewpoint refinement strategy for motion safety and energy-saving. The planner [22], [23] receives the target and generates an executable trajectory to complete the exploration.

The above process is repeated until there is no frontier in space, which signifies the completion of autonomous exploration.

IV. VIEWPOINTS GENERATION FOR FRONTIERS

The frontier clusters are established by an incremental frontier information structure [12]. The frontier information is finely divided into clusters by obstacles. Then, the average position of each frontier cluster P_{fx}, P_{fy}, P_{fz} is obtained.

A. Viewpoints Sampling Generation

In previous work, RRT-like algorithms were used to randomly sample in free space [5], [6], [7], but these methods consume a lot of time and computing resources to generate and keep sparse candidate viewpoints in large space, so they are not suitable for sampling around the frontiers. In addition, the work represented by [8] sampled uniformly near the frontier cluster with a fixed pattern, which does not guarantee the randomness of the sampling and affects the quality of the initial candidate viewpoints.

Inspired by [30], which generates flight corridor fast via a 3D Gaussian sampler in trajectory optimization, this letter proposes a 2D Gaussian sampler for higher-quality candidate viewpoints sampling. In the random sampling method, the sensor horizontal field of view angle α_{fov} , the maximum sensor distance s_{dis} , and the vector $\mathbf{L}_f$ connecting the two farthest frontier points in the same cluster (the size of the cluster) are fully considered.

A 2D Gaussian distribution $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is applied in 3D space by setting the Z-axis coordinates of all sampling points to be the same as P_{fz} . To generate random candidate viewpoints $\mathbf{VP}_c = \{VP_{cx}, VP_{cy}, VP_{cz}\} \in \mathbb{R}^3$, the optimized sampling mean $\boldsymbol{\mu} = \{\mu_x, \mu_y\} \in \mathbb{R}^2$ is set as the point perpendicular to $\mathbf{L}_f$ and 1/2 the length of s_{dis} away from the frontier average in the known area, instead of simply taking the frontier average as the sampling center. The covariance matrix $\boldsymbol{\Sigma}$ can be expressed as:

$$\begin{cases} \boldsymbol{\Sigma} = \mathbf{A}\mathbf{A}^T, \mathbf{A} = \mathbf{R}\mathbf{S} \\ \mathbf{R} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \\ \mathbf{S} = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \end{cases} \quad (1)$$

where $\mathbf{R}$ is the rotation matrix, θ is the clockwise rotation angle of all sampling points in the world coordinate, and $\mathbf{S}$ is the scale matrix representing the degree of stretching or compression compared to the standard one. The matrix parameters θ , s_x , and s_y can be calculated as follows:

$$\theta = 2\pi - \cos^{-1} \frac{\mathbf{L}_f \cdot \mathbf{v}_Y}{\|\mathbf{L}_f\| \cdot \|\mathbf{v}_Y\|} \quad (2)$$

$$\begin{cases} s_x = \sqrt{\frac{s_{dis} \cdot \tan(\alpha_{fov}/2)}{\|\mathbf{L}_f\|}} \\ s_y = \frac{1}{s_x} \end{cases} \quad (3)$$

where $\mathbf{v}_Y$ is the vector parallel to the Y-axis in the world coordinate.

B. Candidate Viewpoint Selection for Each Frontier

After obtaining sufficient random candidate viewpoints, the best candidate viewpoint that covers the most frontier cells needs to be selected to represent the corresponding frontier cluster. Similar to [15], a yaw calculation method is used in this study to determine the yaw angle VP_{yaw} as Algorithm 1 illustrated.

Then, the most coverable candidate viewpoint for the i th frontier cluster $\mathbf{VP}_i = \{VP_{ix}, VP_{iy}, VP_{iz}, VP_{iyaw}\}$ is selected by evaluating the number of frontier cells within the sensor range. The closest viewpoint to the current position of the UAV will be

Algorithm 1: Yaw Calculation Method.

Input: the candidate viewpoints VP_c , the frontier point in the cluster $P_{f1}, P_{f2}, \dots, P_{fn}$

Output: the yaw angle of the candidate viewpoint VP_{yaw}

- 1 **Initial** all_yaw = 0; ref_dir = $(P_{f1} - VP_c).normalized()$;
- 2 **for each** P_{fi} in the Cluster $P_{f1}, P_{f2}, \dots, P_{fn}$ **do**
- 3 dir = $(P_{fi} - VP_c).normalized()$;
- 4 yaw = $\text{acos}(\text{dir} \cdot \text{ref_dir})$;
- 5 all_yaw += yaw;
- 6 **end for**
- 7 $VP_{yaw} = \text{all_yaw}/n + \text{atan2}(\text{ref_dir}[1], \text{ref_dir}[0])$;

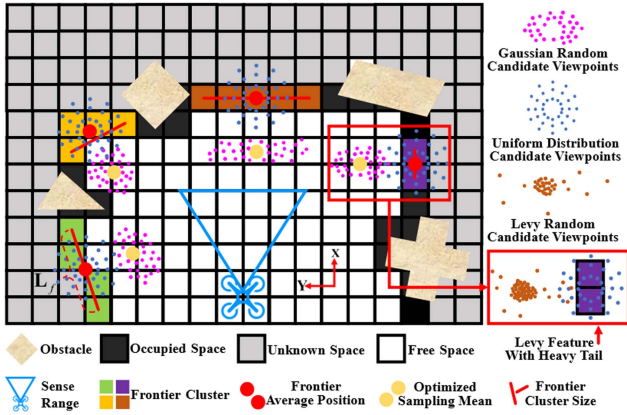


Fig. 3. Performance of three sampling methods (uniform, Gaussian, and Levy) in frontier clusters. The red box demonstrates the characteristics of different methods when sampling in the same frontier cluster.

picked finally if there are more than two candidate viewpoints with the same maximum covered frontier number.

C. Results and Analysis

As it is mentioned before, efficient viewpoint generation methods are the basis for improving exploration results. As shown in Fig. 3, uniform distribution [12] always selects candidate points around the frontier average with the same strategy, and the points falling into unknown regions will be culled. Meanwhile, the point number and interval should be set carefully to balance time and quality. The Levy distribution, another random sampling method with a heavy tail feature, generates most points concentrated in a small area and a few points scattered in a large area. It is highly dependent on the parameter setting and appears extreme and uncontrollable. However, our sampling method can randomly generate more reasonable candidate viewpoints considering the cluster size and the sensor range, which is more flexible for different cases.

The above three methods are tested during the exploration in office room 1 (Fig. 7) in terms of sampling time and viewpoint quality (i.e., the maximum number of covered frontiers at the viewpoint). As illustrated in Fig. 4, the proposed method can generate more efficient initial candidate viewpoints faster, which can catch more frontier voxels (EIG). Meanwhile, due to the sensor range, it is not necessary to go deep into the frontier for

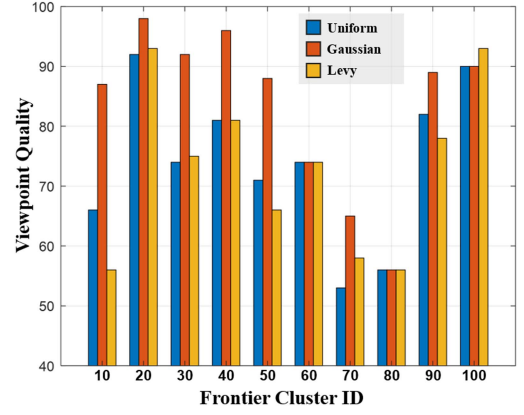


Fig. 4. Comparison results of sampling viewpoint quality. The average sampling time: Uniform, 4.13 ms; Gaussian, 2.31 ms; Levy, 2.18 ms.

coverage. Therefore, the Gaussian sampling which considers the sensor range and frontier cluster size can reduce the inaccurate decision and motion required for exploration.

V. HEURISTIC VIEWPOINT DECISION

Now, a set of candidate viewpoints from each frontier cluster have been selected, and only the next viewpoint as the local target needs to be determined by the heuristic evaluation function.

As mentioned above, [12] modeled the decision problem for the next viewpoint as a complex TSP, attempting to reduce the effect of local optimum by considering all viewpoints and planning a global tour path. Better performance was achieved in a self-designed small simulation map. However, recent work[4] questioned this method and employed a simple and greedy exploration heuristic. It performs on par with the state-of-the-art methods in smaller environments and outperforms them in larger environments, both in terms of exploration speed and computational efficiency. Based on our previous research experience, simple heuristics are indeed more effective than complex approaches that consider the connection between viewpoints.

This study adopts the heuristic method for two reasons. One is to plan a global tour, and only taking the first point in the tour as the local target is not meaningful. This is because the high-frequency change and update of the surrounding local environment may make the UAV fail to reach the current viewpoint when the information is updated, and the UAV has to go to a new viewpoint. The other is to meet the coverage requirements so that the unknown space will be completely visited with more concerns on the current immediate information and the order of visits.

Therefore, this letter designs a heuristic evaluation function to improve the current immediate information gain and avoid unnecessary repeated exploration. Meanwhile, a viewpoint refinement strategy is employed for safety and energy-saving.

A. Heuristic Evaluation Function Design

1) *Boundary Cost*: Exploration space has its boundary, and an imperfect viewpoint decision strategy makes it more expensive for the quadrotor to thoroughly explore boundary

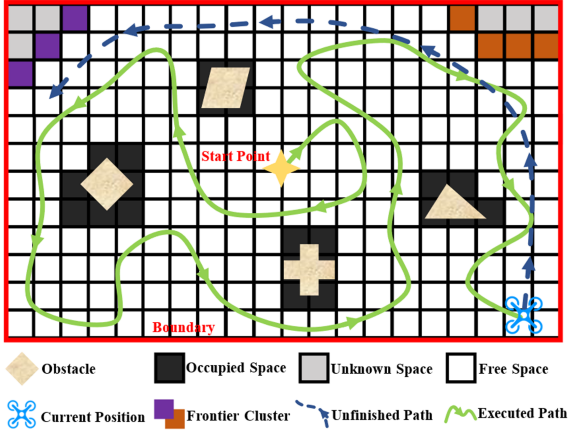


Fig. 5. Repeated visits to the boundary.

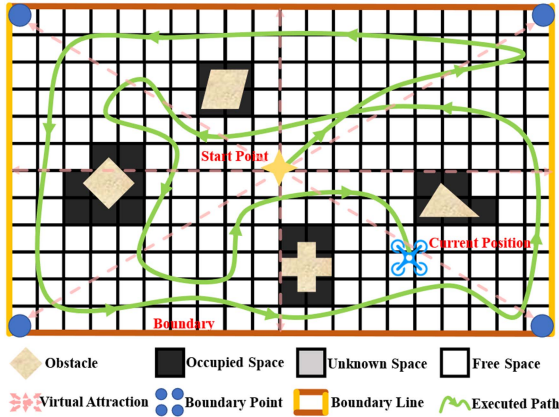


Fig. 6. Exploration with boundary cost.

areas. As shown in Fig. 5, this is reflected in the fact that unknown environment information is still left behind by the UAV when it first reaches the boundary areas after a long journey. This makes it necessary to repeatedly visit the boundary at a higher cost after completing the exploration of other space, especially in a large-scale environment.

Therefore, a fresh boundary cost is carefully proposed to avoid expensive boundary exploration costs, and to the best of our knowledge, this study is the first to design such a cost. Before the exploration begins, the space to be explored represented by four corner points has been artificially defined. The boundary cost c_b includes the corner point cost c_{bp} and the connection line cost c_{bl} , and it is calculated as follows:

$$c_b = \lambda_{bp}c_{bp} + \lambda_{bl}c_{bl} \quad (4)$$

where c_{bp} is the minimum distance between the corner point and the candidate viewpoint $\mathbf{VP}_i$, c_{bl} is the minimum vertical distance from the connection line to $\mathbf{VP}_i$, and λ_{bp} , λ_{bl} are responding positive weights.

With the boundary cost, as shown in Fig. 6, the UAV will first focus on the boundary areas under the influence of virtual attraction and avoid unnecessary long-distance repeated exploration.

2) *Environment Structure Cost*: In a complex environment, the less thorough access to environment information, the greater the probability of repeated exploration. Especially in a cluttered environment, narrow and complex corners around the obstacle often need extra attention. Therefore, this letter proposes an environment structure cost that is represented by the mean distance between the viewpoint and obstacles. The environment structure cost c_e can be represented as:

$$c_e = \frac{\sum_{i=1}^n \|\mathbf{P}_i - \mathbf{P}_f\|_2}{n} \quad (5)$$

where $\mathbf{P}_i$ is the position of the sampling viewpoints in Section IV-A, which are in obstacles, and n is the number of points above.

3) *Direction Change Cost*: For motion consistency, the direction change cost is proposed to evaluate the change in motion direction. Here, the direction change cost c_d is defined as:

$$c_d = \cos^{-1} \frac{(\mathbf{P}_{uav} - \mathbf{VP}_i) \cdot \mathbf{v}_{uav}}{\|\mathbf{P}_{uav} - \mathbf{VP}_i\| \cdot \|\mathbf{v}_{uav}\|} \quad (6)$$

where $\mathbf{v}_{uav}$ is the current velocity of the quadrotor.

4) *Distance Cost*: Distance cost is adopted in this study to act as a tiebreaker. Inspired by [12], the path searched by the A* algorithm is treated as the real distance between the current position $\mathbf{P}_{uav}$ and the viewpoint $\mathbf{VP}_i$. Compared with the traditional method using Euclidean distance, using the path length of A* in autonomous exploration with negligible computational cost is more efficient.

$$c_{dis} = \text{AstarPathLength}(\mathbf{P}_{uav}, \mathbf{VP}_i) \quad (7)$$

The final total cost of the i th candidate viewpoint is expressed as:

$$c_{total,i} = \lambda_b c_b + \lambda_e c_e + \lambda_d c_d + \lambda_{dis} c_{dis} \quad (8)$$

where λ_b , λ_e , λ_d , and λ_{dis} are positive weights. The one with the lowest cost is temporarily taken as the next viewpoint before the refined check.

B. Viewpoint Refine

The position $\mathbf{VP}_{next} \in \mathbb{R}^3$ and the yaw ψ_{next} of the next viewpoint are determined after the sampling and evaluation. However, even if the viewpoint picked after the evaluation is numerically optimal, in the actual exploration process, position motions with short distances are generally non-essential for complete EIG within sensor limits. Besides, affected by the environment structure cost which tends to drive UAVs to explore complex corners, the planning with a short distance but a rapid yaw change lacks perception and is easy to cause a collision with obstacles.

To this end, this letter proposes a viewpoint refinement strategy for energy-saving and safety. When the distance between the current position $\mathbf{P}_{uav}$ and the next viewpoint $\mathbf{VP}_{next}$, which is calculated by A*, is less than the threshold D_{ref} , the case is triggered. The final target position $\mathbf{P}_{target} \in \mathbb{R}^3$ is refined with the current position $\mathbf{P}_{uav}$, and the final target yaw ψ_{target} is refined with the yaw calculated in the same way at $\mathbf{P}_{uav}$ in Section IV-B. In this way, the EIG corresponding to

Algorithm 2: Viewpoint Refinement Strategy.

Input: the current position $\mathbf{P}_{uav}$, the next viewpoint position $\mathbf{VP}_{next}$, the yaw ψ_{next} , and the threshold D_{ref}
Output: final target position $\mathbf{P}_{target}$; final target yaw ψ_{target}

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1 Initialize  $\mathbf{P}_{uav}$ ,  $\mathbf{VP}_{next} = \text{GetOriginalVP}()$ ;
2  $\text{AstarDistance} = \text{CalAstarPathLength}(\mathbf{P}_{uav}, \mathbf{VP}_{next})$ ;
3 If ( $\text{AstarDistance} < D_{ref}$ ) then
4    $\mathbf{P}_{target} = \mathbf{P}_{uav}$ ;
5    $\psi_{target} = \text{UpdateNewYaw}(\mathbf{P}_{uav})$ ;
6 else
7    $\mathbf{P}_{target} = \mathbf{VP}_{next}$ ;  $\psi_{target} = \psi_{next}$ ;

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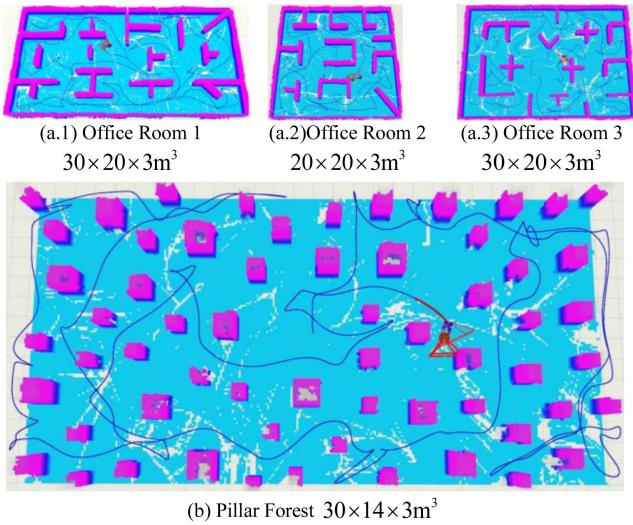


Fig. 7. Simulation Maps.

$\mathbf{VP}_{next}$ can also be completed by ψ_{target} at $\mathbf{P}_{uav}$ because there will always be potential equivalent viewpoints in such a close case. Simulations and extensive real-world tests demonstrate the effectiveness of the strategy in safety and energy-saving. More details can be seen in Algorithm 2 and the attached video.

VI. EXPERIMENTS AND RESULTS

A. Simulations

As shown in Fig. 7, rich scenarios including three office rooms and a pillar forest were set to validate our method by comparing it with the classic frontier-based autonomous method FUEL [12]. To verify the scalability of our method in almost all scenarios and to achieve excellent exploration results, this study used the same simulation maps provided by FUEL.

In all the simulations, the dynamic motion limits were set as $v_{max} = 2.0$ m/s, $a_{max} = 3.0$ m/s², and $\xi_{max} = 0.9$ rad/s. The heuristic function weights were set as $\lambda_b = 0.4$, $\lambda_e = 0.3$, $\lambda_d = 0.2$, and $\lambda_{dis} = 0.1$. The horizontal view field of the sensor α_{fov} was set as 60 deg with a maximum range s_{dis} of 4.5 m. For both

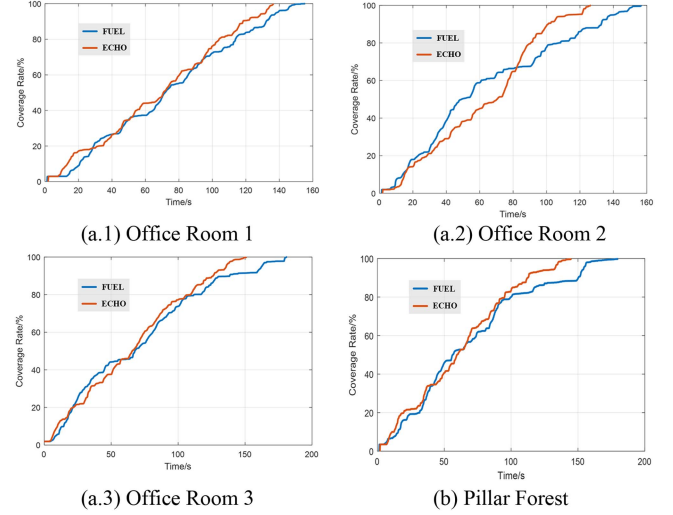


Fig. 8 Exploration process (coverage with time) comparison of FUEL and ECHO in simulations.

TABLE I
EXPLORATION RESULTS OF SIMULATIONS

Scene	Method	Exploration Time /s			Exploration Path /m		
		Ave	Max	Min	Ave	Max	Min
Office Room	1 FUEL	168	176	146	223	248	211
	1 ECHO	123	147	118	197	208	189
	2 FUEL	140	148	134	189	200	176
	2 ECHO	112	131	105	163	189	150
	3 FUEL	168	180	159	228	245	217
	3 ECHO	143	157	131	195	216	191
Pillar Forest	FUEL	155	166	149	210	220	199
	ECHO	140	163	131	196	217	182

methods with the same initial configuration, each scenario was tested with the same computational device over 10 times.

1) *Office Room*: Three different cluttered office scenarios were used to verify the effectiveness of the method in a typical indoor environment. During exploration, the quadrotor first visited the boundary space and paid extra attention to the complex corners around the obstacles. As shown in Fig. 8, ECHO can finish the exploration with full coverage at an almost constant undecreasing rate. The results in Table I demonstrates that our method can improve exploration efficiency by 15–25% compared to FUEL in all three indoor office scenarios.

2) *Pillar Forest*: In a forest with uniformly distributed obstacles, our method can also finish a thorough exploration of the local space before visiting other spaces. One of the key requirements for exploration is complete coverage, which indicates that it is better to obtain EIG at a stable linear speed and end up at a small cost to make up for the missed part of EIG like FUEL in Fig. 8. The results in Table I indicate that our method can catch the EIG efficiently and stably in the overall exploration process compared to FUEL.

B. Real-World Exploration Tests

There are more random disturbances in the real world, posing greater challenges to the implementation of the proposed method

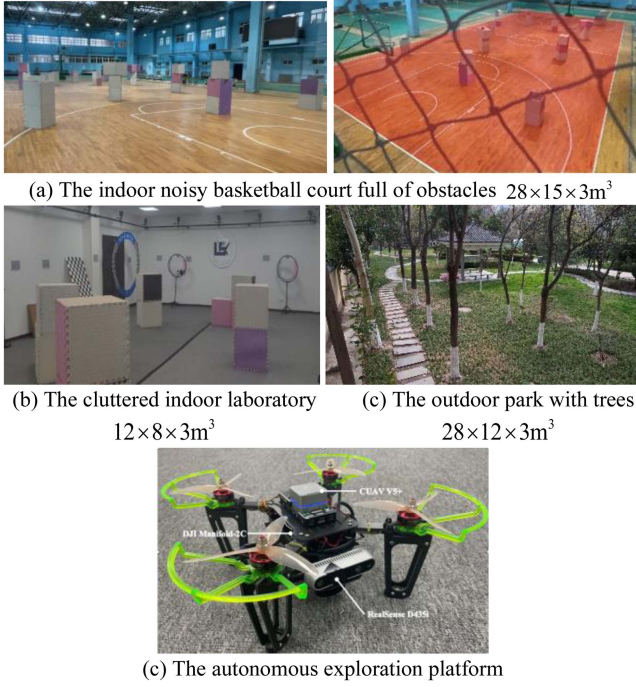


Fig. 9. Conditions of real-world tests.

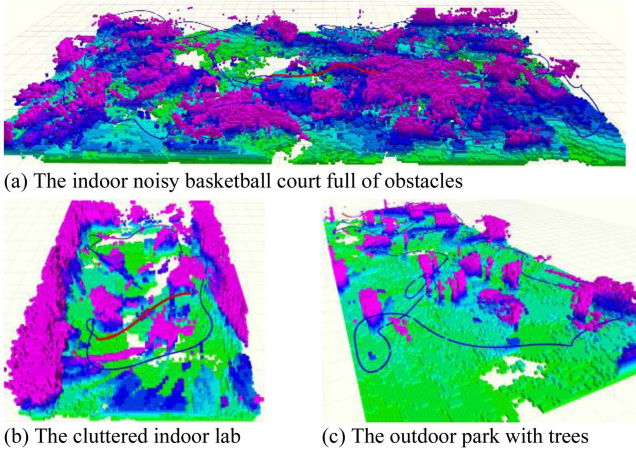


Fig. 10. Exploration results in real-world tests. More details can be seen in the video.

ECHO. To further validate the practicality and stability of our method, real-world tests were conducted by our self-developed onboard platform in the indoor basketball court, indoor clutter laboratory, and outdoor park, as shown in Fig. 9. The above sites represent the common application scenarios for autonomous exploration. The dynamic motion limits were set as $v_{\max} = 1.5 \text{ m/s}$, $a_{\max} = 2.0 \text{ m/s}^2$ in the large-scale basketball court tests and $v_{\max} = 1.0 \text{ m/s}$, $a_{\max} = 1.2 \text{ m/s}^2$ in the cluttered laboratory and park. All other settings were the same as those in the simulations.

In real-world tests, all the modules including state estimation [30], planning, and control [32] were run onboard on an Intel i7-8550U processor and a real-sense camera without any external device. Both our method ECHO and the FUEL method were

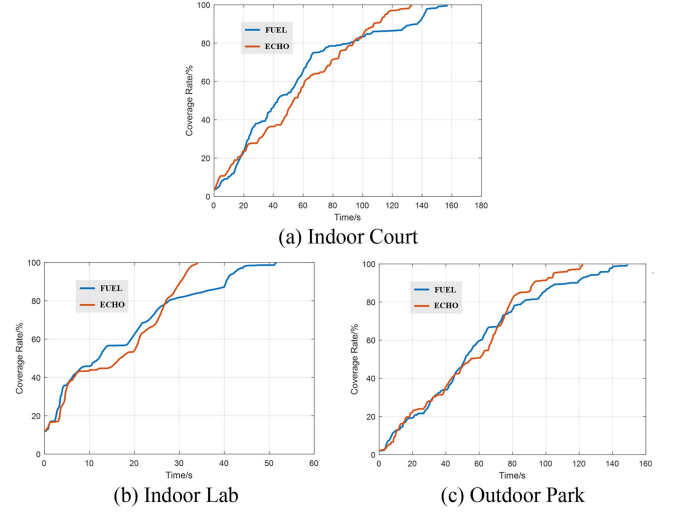


Fig. 11. Exploration process in real-world tests.

TABLE II
EXPLORATION RESULTS OF REAL-WORLD TESTS

Scene	Method	Exploration Time/ s			Exploration Path/ m		
		Ave	Max	Min	Ave	Max	Min
Indoor Court	FUEL	162	168	147	171	180	154
	ECHO	132	144	120	145	165	128
Indoor Lab	FUEL	48	61	43	62	75	55
	ECHO	38	46	31	53	59	46
Outdoor Park	FUEL	153	164	141	179	187	163
	ECHO	121	133	117	147	159	134

tested at least five times under the same configuration to validate the stable performance and security. The platform was required to autonomously gain complete environment information as fast as possible and safely.

During almost all the real-world tests, the FUEL method using TSP did not solve the local optimum problem. It determined the next viewpoint aimlessly and left the frontiers near the visited spaces, leading to a long journey back to the former spaces for a second exploration. As shown in Fig. 11, FUEL experiences a significant slowdown of efficiency in the latter half of the exploration process, which is contradictory to the demand for efficient and complete EIG. However, attributed to the more efficient viewpoint determination method with full consideration of the EIG, our method ECHO thoroughly explored the visited spaces, avoided repeated work, and completed exploration with a steady efficiency. Table II demonstrates that our method can steadily improve exploration efficiency in multiple validations.

VII. CONCLUSION

In this letter, an efficient heuristic viewpoint determination method is proposed for frontier-based autonomous exploration. The proposed method consists of viewpoint generation, selection, and refinement. As one of the most important parts of the autonomous exploration framework, the viewpoint determination method can reduce repeated and unnecessary exploration and enable a significant improvement of exploration time and path with stable efficiency. Our method is validated in different

simulations and real-world tests, and it can improve exploration efficiency by 15%–25% compared to the state-of-the-art method.

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